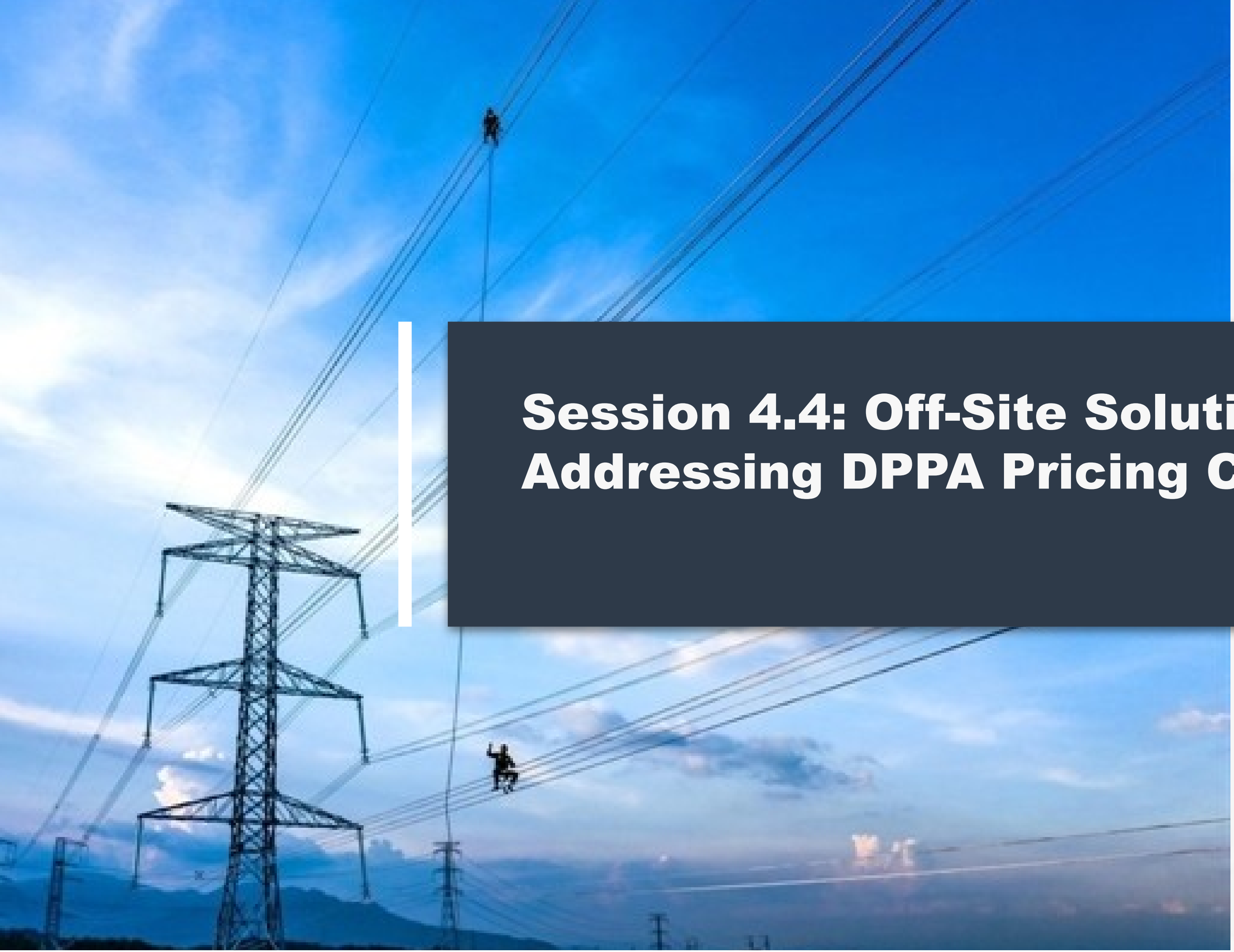


Session 4.4: Off-Site Solutions: Addressing DPPA Pricing Concerns

About the 2026 Update

Current repo: CFO calculator, tested settlement engine, formula docs, and workshop readiness analysis

35 tests passing	**3** factory profiles
5 pricing inputs	**0** critical gaps

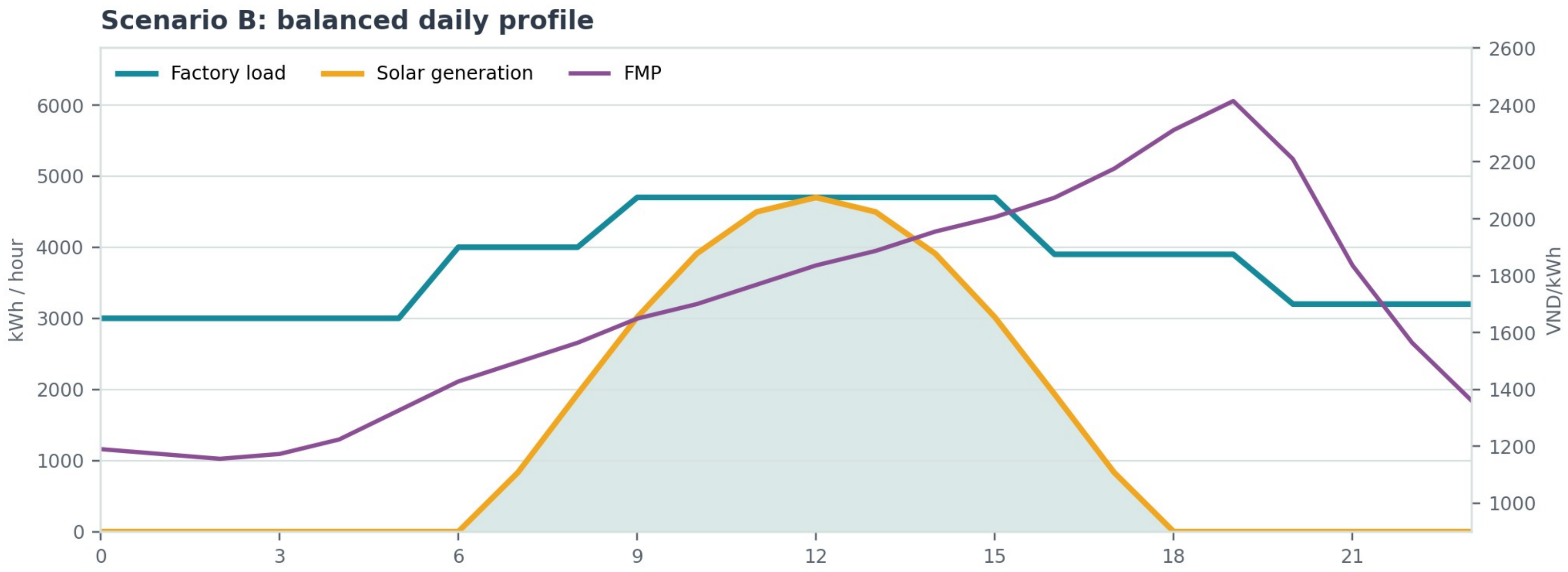
The reference deck explained why DPPA buyers worry about market price. This update uses the current repository to show the actual settlement mechanics behind that concern.

Updated storyline:
- FMP and CFMP remain visible buyer concerns
- Hourly matched quantity is the cost driver
- The CfD cancels much of the FMP exposure on matched kWh
- Settlement quantity mode determines overgeneration risk

Current app status: 35 tests passing, three synthetic profiles, default matched-mode settlement.

Understanding Market Price

Spot market price remains the visible buyer concern, but the current calculator shows where it truly affects cost



Understanding Market Price

Matched quantity, not headline FMP, drives the buyer's DPPA cost exposure

2,100
strike price Pc

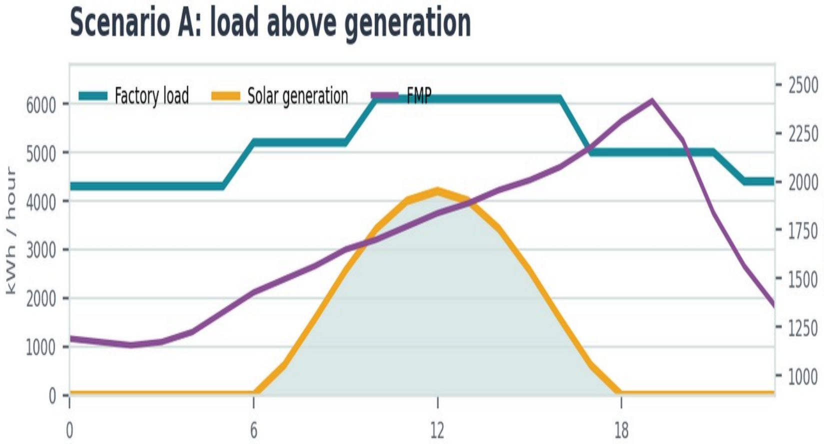
523.34
DPPA charge

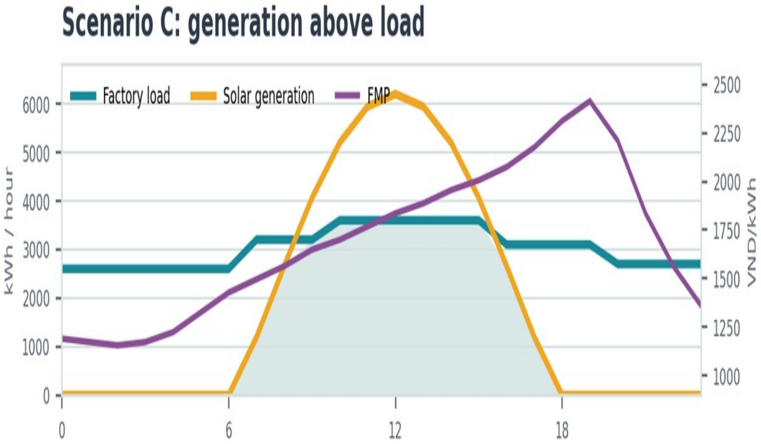
1.027263
Kpp loss factor

FMP remains visible in the EVN market charge and developer CfD.
Matched kWh mostly collapse toward strike price plus DPPA charge.
Shortfall and excess volumes decide whether that clean story holds.

Understanding Market Price

The repo uses a synthetic daily FMP curve to teach price sensitivity without hiding the settlement logic.





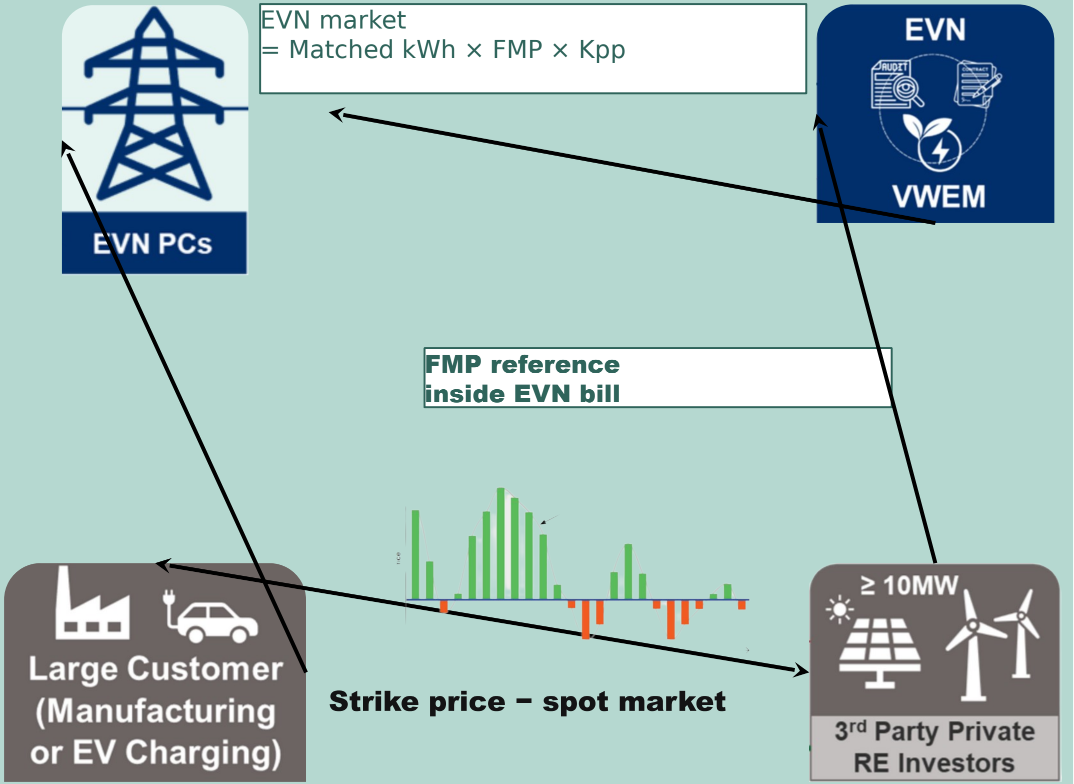
Payment Mechanisms

Buyer cost = EVN market + DPPA charge + retail shortfall + developer CfD

Developer CfD
= Contract quantity × (Strike − FMP)

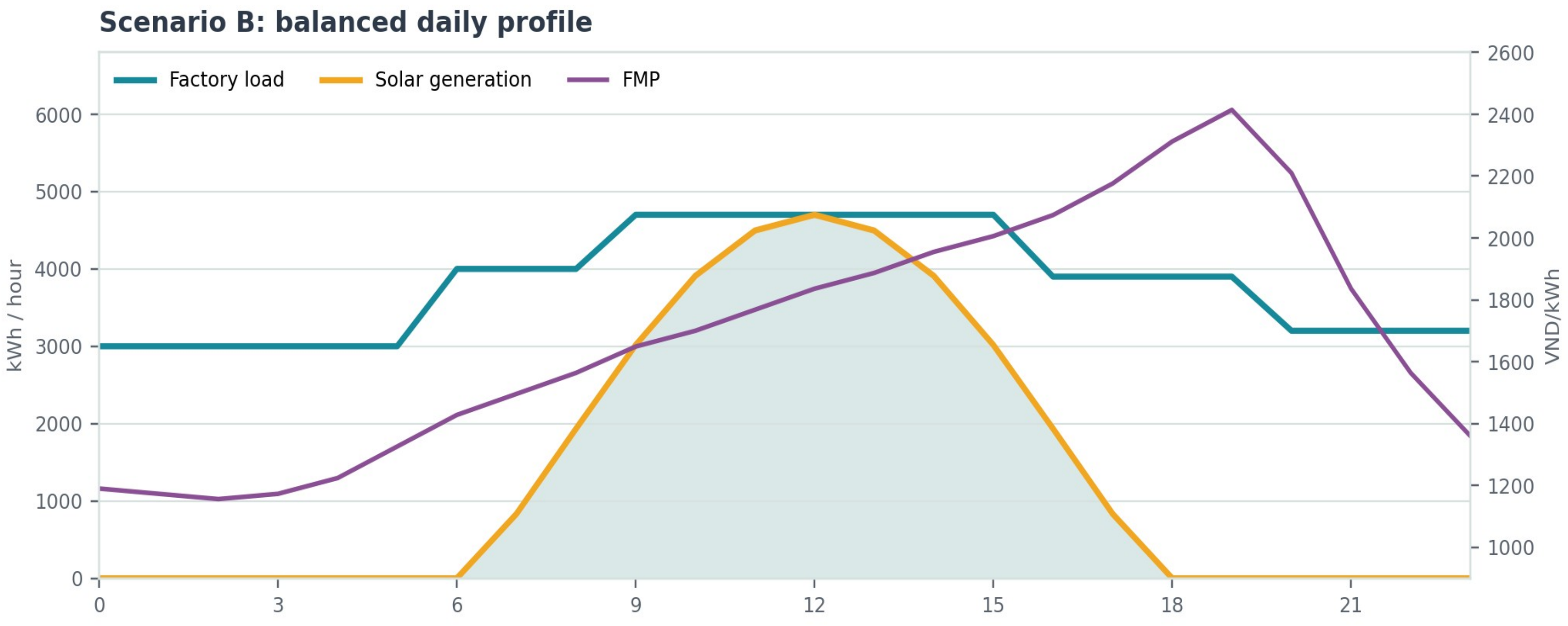
DPPA service charge
= Matched kWh × C_DPPA

Retail shortfall
= (Load − matched kWh) × retail tariff



Diurnal Profile

Accurate allocation of QKHhc is critical for determining DPPA cost settlement



QKHhc is the key factor in all DPPA cost calculations: buyers must calculate it accurately.

Matched volume

QKHhc(i) = min(QKH(i), Qm(i))

Shortfall

max(QKH(i) − Qm(i), 0)

Excess

max(Qm(i) − QKH(i), 0)

Matched mode: contract quantity equals matched energy.

Generation mode: contract quantity equals renewable generation, so excess solar can still create CfD exposure.

Allocated mode: contract quantity follows the allocation in the contract.

Buyer implication: negotiate the quantity definition before debating a headline strike price.

Payment Mechanism under the CFD

Strike price determines financial settlement through CFD, converting market price volatility into predictable financial flows

EVN Market + DPPA Charge + CfD

$= Q \times FMP + Q \times C\_DPPA + Q \times (Strike - FMP)$

$= Q \times Strike + Q \times C\_DPPA$ (+ loss adjustment)

Pc(i): committed strike price at cycle i.

FMP(i): hourly market reference price.

Qca(i): contract quantity; default app mode equals matched volume.

Quantify DPPA Costs to Assess Feasibility & Negotiate RE Supply

Feasibility analysis enables buyers to compare DPPA against traditional procurement with confidence

Metric	Load > Gen	Load = Gen	Load < Gen
Daily load kWh	123,100	91,300	72,600
Matched kWh	28,552	33,080	32,508
Shortfall kWh	94,548	58,220	40,092
Excess kWh	0	0	11,814
DPPA cost VND	274,875,485	210,690,815	171,094,487
BAU cost VND	258,510,000	191,730,000	152,460,000
Savings vs BAU	**-16,365,485**	**-18,960,815**	**-18,634,487**

Session 4.5: Preparing for DPPA Implementation

allotrope PARTNERS

CEBA Clean Energy Buyers Association

Interactive Exercise:
DPPA Scenario Analysis (~60 minutes)

Use the CFO calculator to test one factory load profile against three questions:

- How much load is matched by renewable generation?
- Which hours create retail shortfall exposure?
- Which settlement quantity mode is acceptable before negotiating strike price?

Start with the balanced profile, then compare Load > Gen and Load < Gen.

Scenario DPPA Negotiation

Your factory is located in Northern Vietnam, within an industrial park, and is currently purchasing electricity from an EVN-subsidiary or industrial park electricity retailer. Annual demand is 150 GWh.

Determine the essential conditions for proceeding with an appropriate DPPA arrangement:

- Electricity retailer
- Optimal renewable energy source
- Required capacity and load match
- DPPA contract duration
- Settlement quantity mode
  - Risk allocation for shortfall and excess